AI Application Lecture 16

3D Rendering Objective

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[a]I thank my collaborator, Jing WANG, for cooperating in creating the teaching materials.

Outline

1. Introduction

2. Preliminaries: Mathematical Notations

3. What Is the Goal? – Triangle Mesh

4. Adapter from a Latent Diffusion Model – Implicit 3D Field and Surface Extractor

5. The Complete 3D Rendering AI Pipeline – Hunyuan3D as an Example

6. Summary

From image generation to 3D generation

In image generation, the goal is to generate a 3 or 4-channel image. We can change the resolution by changing the noise generation with help of variable-shape architecture, i.e., attention, convolutional layer, RoPE.

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In image generation, the goal is to generate a 3 or 4-channel image. We can change the resolution by changing the noise generation with help of variable-shape architecture, i.e., attention, convolutional layer, RoPE.

What is the goal of 3D rendering, and how can we make the resolution adaptable to applications?

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At the end of this lecture, you will be able to

- **Formulate** the goal of 3D rendering.
- **Construct** a scheme to adapt resolutions to application situations in 3D rendering. **Explain** the strategy of 3D rendering AI.

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Preliminaries: mathematical notations

Set:

- Sets: $\mathcal{A}$
- Membership: $x \in \mathcal{A}$
- Empty set: $\{\}$
- Roster notation: $\{a, b, c\}$
- Set-builder: $\{x \in \mathcal{A} | P(x)\}$
- Cardinality: $|\mathcal{A}|$
- Real numbers: $\mathbb{R}, \mathbb{R}_{>0}, \mathbb{R}_{\geq 0}$
- Integers: $\mathbb{Z}, \mathbb{Z}_{>0}, \mathbb{Z}_{\geq 0}$
- Integer range: $[1, k]_{\mathbb{Z}} \coloneqq \{1, \ldots, k\}$

Function:

- $f : \mathcal{X} \to \mathcal{Y}$: f maps from $\mathcal{X}$ to $\mathcal{Y}$.
- $y = f(x)$: The output of f for input x.

Definition:

- $(\mathrm{LHS}) \coloneqq (\mathrm{RHS})$: Left side is defined by the right side.

Preliminaries: mathematical notations

Sequence: Denoted by $\boldsymbol{a} = (a_1, a_2, \dots)$.

- A function $\boldsymbol{a} : [1, n]_{\mathbb{Z}} \to \mathcal{A}$.
- Length is denoted by $|\boldsymbol{a}|$.

Vector: Denoted by $\boldsymbol{v}$.

- A column of numbers, $\boldsymbol{v} \in \mathbb{R}^n$.
- i-th element is v_i.

Matrix: Denoted by $\boldsymbol{A}$.

- $m \times n$ matrix: $\boldsymbol{A} \in \mathbb{R}^{m,n}$.
- (i, j)-th element is $a_{i,j}$.
- Transpose: $\boldsymbol{A}^\top$.

Tensor: Denoted by $\underline{\boldsymbol{A}}$.

- Simply a multi-dimensional array.
- Vector $\to$ 1st-order, Matrix $\to$ 2nd-order.

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Question: What is the goal in 3D rendering?

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In image generation, our objective was to output a 3rd-order tensor, i.e., three matrices corresponding to red channel, green channel, and blue channel.

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A common representation of a renderable 3D surface is a **triangle mesh** (e.g., `Trimesh` in Python, GLB file).

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- The face matrix $\boldsymbol{F} \in \{0, 1, ..., N_V - 1\}^{N_F,3}$ stores three vertex indices for each triangular face.

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Here, the ith row $\begin{bmatrix} f_{i,0} & f_{i,1} & f_{i,2} \end{bmatrix}$ of $\boldsymbol{F}$ indicates the three vertex indices of the ith triangle.

In other words, the three vertices of the ith triangle face is $\boldsymbol{v}_{f_{i,0}}, \boldsymbol{v}_{f_{i,1}}, \boldsymbol{v}_{f_{i,2}}$.

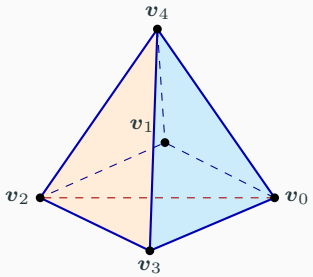
Pyramid-shape example

Example (Five vertices and six triangular faces)

We can construct a pyramid-shape mesh as follows:

$$\boldsymbol{V} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \boldsymbol{F} = \begin{pmatrix} 0 & 2 & 1 \\ 0 & 3 & 2 \\ 0 & 1 & 4 \\ 1 & 2 & 4 \\ 2 & 3 & 4 \\ 3 & 0 & 4 \end{pmatrix}.$$

The first 4 rows of $\boldsymbol{V}$ are vertices of the square base. The first 2 rows of $\boldsymbol{F}$ form the square base, divided into two right isosceles triangles. The other four rows form the lateral faces.



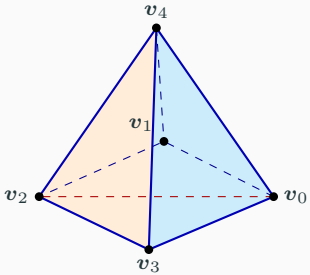
Pyramid-shape example

Exercise (Four triangular lateral faces)

Consider the same square pyramid as in the preceding example, with the same five vertices and vertex matrix $\boldsymbol{V}$.

Remove the two triangles forming the square base, retaining only the four triangular lateral faces.

Write the resulting face matrix $\boldsymbol{F} \in \{0, 1, 2, 3, 4\}^{4,3}$ using zero-based vertex indices.



Pyramid-shape example

Answer:

$$\boldsymbol{F} = \begin{pmatrix} 0 & 1 & 4 \\ 1 & 2 & 4 \\ 2 & 3 & 4 \\ 3 & 0 & 4 \end{pmatrix}.$$

Each row specifies one lateral triangle, and the two base triangles $(0, 2, 1)$ and $(0, 3, 2)$ are excluded. The vertex matrix $\boldsymbol{V}$ is unchanged from the preceding example.

This answer also shows that $\boldsymbol{V}$ does not uniquely determine a triangle mesh.

Mesh generation — Answer

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Answer: The objective is to generate a set of 3D faces. They are represented as a vertex matrix and face matrix.

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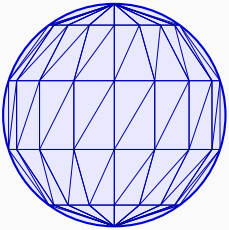
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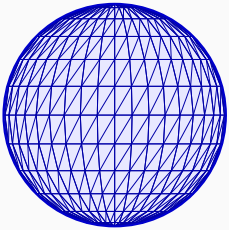
Question: The required resolution of mesh depends on applications and may change at the inference phase. How

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The resolution may be determined at the inference phase.

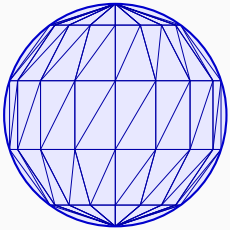


Coarse triangle mesh

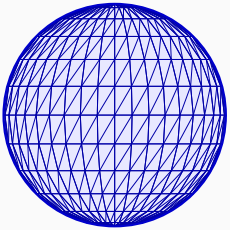


Fine triangle mesh

The resolution may be determined at the inference phase.



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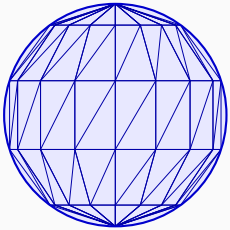


Fine triangle mesh

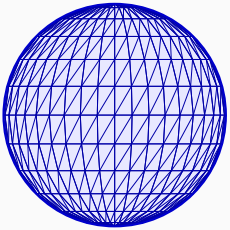
The required resolution depends on applications (e.g., 300 faces in the 2 player mode, 200 faces in the 4 player mode.). We design a pipeline so that one NN can generate 3D mesh at arbitrary resolutions.

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Coarse triangle mesh



Fine triangle mesh

The required resolution depends on applications (e.g., 300 faces in the 2 player mode, 200 faces in the 4 player mode.). We design a pipeline so that one NN can generate 3D mesh at arbitrary resolutions.

How can we realize adaptable resolution mesh generation?

Solution: generating a scalar field

In a 3D rendering AI pipeline, a NN generates a scalar **3D field** instead of directly generating mesh.

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In a 3D rendering AI pipeline, a NN generates a scalar **3D field** instead of directly generating mesh.

Here, a 3D field is a scalar function $f : \mathbb{R}^3 \to \mathbb{R}$ such that the following hopefully holds:

$$\begin{cases} f(\boldsymbol{q}) \leq \ell & \text{if } \boldsymbol{q} \text{ is inside the object,} \\ f(\boldsymbol{q}) > \ell & \text{if } \boldsymbol{q} \text{ is outside the object.} \end{cases}$$

Here, $\ell \in \mathbb{R}$ is called the **extraction level**.

Outline of mesh generation from a scalar field

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- Locate contour points (intersections with the level set) on 3D grid edges.
- Construct 3D mesh by connecting the contour points.

Mesh generation — 2D analogy

Since it isn’t easy to imagine 3D from the beginning, let’s consider a 2D case first.

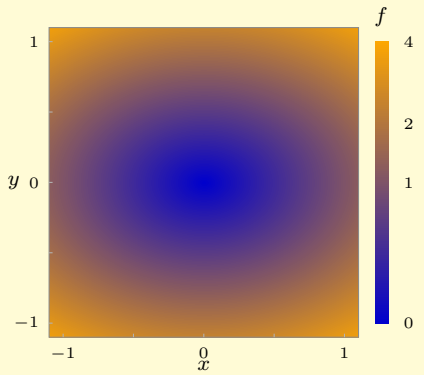
Example: a scalar field in two dimensions

Example (2D Field values in the plane)

Let $f : \mathbb{R}^2 \to \mathbb{R}, \quad f(x, y) = x^2 + 2y^2$.
For example,

$$f(0,0) = 0, \quad f(1,0) = 1, \quad f(0,1) = 2.$$

The diagram shows f on $[-1.1, 1.1]^2$;
the color bar gives the scalar values.



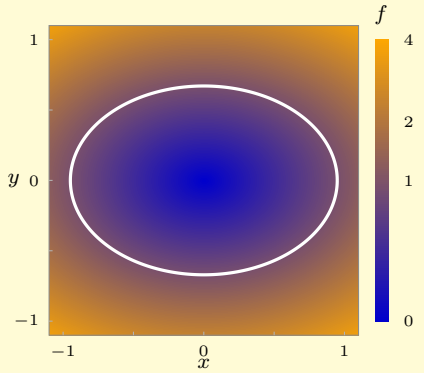
Example: a level set in two dimensions

Example (A contour defined by an equation)

Choose the extraction level $\ell = \frac{9}{10}$. The corresponding level set is

$$\mathcal{S}_\ell = \left\{ (x, y) \in \mathbb{R}^2 \;\middle|\; x^2 + 2y^2 = \frac{9}{10} \right\}.$$

The white curve consists of exactly those points where $f(x, y) = \ell$.



Example: sampling a two-dimensional field

Example (Field values on a regular grid)

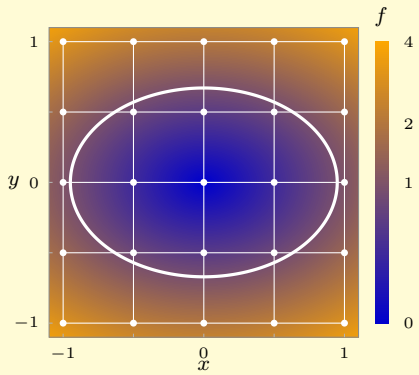
Let $i, j \in [0, 4]_{\mathbb{Z}}$ and set

$$\boldsymbol{q}_{ij} = \left(-1 + \frac{i}{2}, -1 + \frac{j}{2}\right), \qquad G_{ij} = f(\boldsymbol{q}_{ij}).$$

The 5×5 array $\boldsymbol{G} = (G_{ij})$ is

$$\boldsymbol{G} = \begin{pmatrix} 3 & \frac{3}{2} & 1 & \frac{3}{2} & 3 \\ \frac{9}{4} & \frac{3}{4} & \frac{1}{4} & \frac{3}{4} & \frac{9}{4} \\ 2 & \frac{1}{2} & 0 & \frac{1}{2} & 2 \\ \frac{9}{4} & \frac{3}{4} & \frac{1}{4} & \frac{3}{4} & \frac{9}{4} \\ 3 & \frac{3}{2} & 1 & \frac{3}{2} & 3 \end{pmatrix}.$$

The dots mark the 25 grid points; the white curve is the level set $f = \frac{9}{10}$.



Example: locating a contour point on a grid edge

Example (Linear interpolation)

Consider the edge from $\boldsymbol{a} = (\frac{1}{2}, 0)$ to $\boldsymbol{b} = (1, 0)$. Its endpoint values are

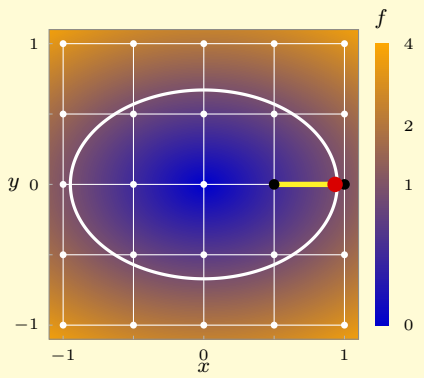
$$f(\boldsymbol{a}) = \frac{1}{4} < \frac{9}{10}, \qquad f(\boldsymbol{b}) = 1 > \frac{9}{10}.$$

Linear interpolation uses

$$t = \frac{\ell - f(\boldsymbol{a})}{f(\boldsymbol{b}) - f(\boldsymbol{a})} = \frac{\frac{9}{10} - \frac{1}{4}}{1 - \frac{1}{4}} = \frac{13}{15}.$$

The extracted point is

$$\boldsymbol{p} = (1-t)\boldsymbol{a} + t\boldsymbol{b} = \left(\frac{14}{15}, 0\right).$$



Example: a line-segment mesh in two dimensions

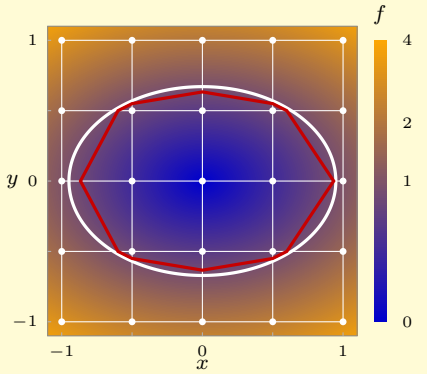
Example (Vertices and connectivity of the extracted contour)

Let $\boldsymbol{V} \in \mathbb{R}^{12\times 2}$ have rows $\boldsymbol{p}_1, \ldots, \boldsymbol{p}_{12}$, where

$$\begin{aligned}
\boldsymbol{p}_1 &= (0, \tfrac{19}{30}), & \boldsymbol{p}_2 &= (\tfrac{1}{2}, \tfrac{11}{20}),\\
\boldsymbol{p}_3 &= (\tfrac{3}{5}, \tfrac{1}{2}), & \boldsymbol{p}_4 &= (\tfrac{14}{15}, 0),\\
\boldsymbol{p}_5 &= (\tfrac{3}{5}, -\tfrac{1}{2}), & \boldsymbol{p}_6 &= (\tfrac{1}{2}, -\tfrac{11}{20}),\\
\boldsymbol{p}_7 &= (0, -\tfrac{19}{30}), & \boldsymbol{p}_8 &= (-\tfrac{1}{2}, -\tfrac{11}{20}),\\
\boldsymbol{p}_9 &= (-\tfrac{3}{5}, -\tfrac{1}{2}), & \boldsymbol{p}_{10} &= (-\tfrac{14}{15}, 0),\\
\boldsymbol{p}_{11} &= (-\tfrac{3}{5}, \tfrac{1}{2}), & \boldsymbol{p}_{12} &= (-\tfrac{1}{2}, \tfrac{11}{20}).
\end{aligned}$$

Let $\boldsymbol{F} \in [1, 12]_{\mathbb{Z}}^{12\times 2}$ have rows $(1,2), (2,3), \ldots, (11,12), (12,1)$. Each row (i, j) represents the line segment

$$\{(1-t)\boldsymbol{p}_i + t\boldsymbol{p}_j \mid 0 \le t \le 1\}.$$



3D field

Similarly, we can generate a 3D mesh from 3D field $f : \mathbb{R}^3 \to \mathbb{R}$.

A surface is represented as a level set

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The surface at level ℓ is

$$\mathcal{S}_\ell \coloneqq \left\{ \boldsymbol{q} \in \mathbb{R}^3 \;\middle|\; f(\boldsymbol{q}) = \ell \right\}.$$

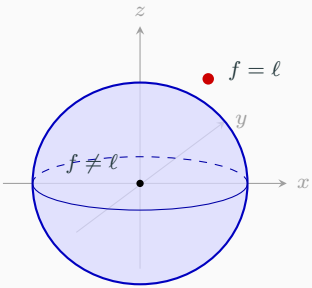
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$$\mathcal{S}_\ell \coloneqq \left\{ \boldsymbol{q} \in \mathbb{R}^3 \;\middle|\; f(\boldsymbol{q}) = \ell \right\}.$$

Thus the surface is specified by an equation rather than by an explicit list of triangles.



Example: a surface extracted from a 3D field

Example (An ellipsoidal level surface)

Consider the scalar field $f(x, y, z) = x^2 + y^2 + z^2$.

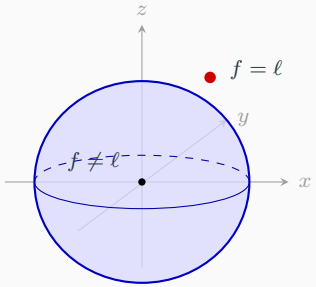
At extraction level $\ell = 1$, the surface is

$$\mathcal{S}_1 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}.$$

For instance,

$$f(0, 0, 0) = 0, \qquad f(1, 0, 0) = 1, \qquad f(2, 0, 0) = 4.$$

Consequently, $(1, 0, 0)$ belongs to $\mathcal{S}_1$, whereas $(0, 0, 0)$ or $(2, 0, 0)$ does not.



Sampling the field on a finite grid

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For $i, j, k \in [0, R]_{\mathbb{Z}}$, define

$$\boldsymbol{q}_{ijk} \coloneqq \boldsymbol{b}_{\min} + \frac{(i, j, k)}{R} \odot (\boldsymbol{b}_{\max} - \boldsymbol{b}_{\min}).$$

Sampling the field on a finite grid

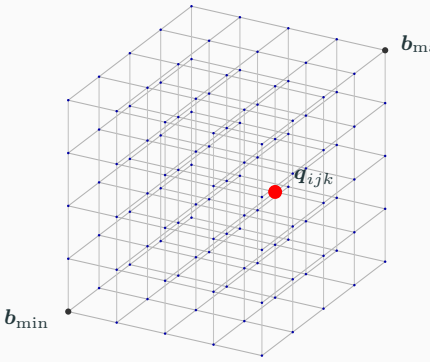
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$$\boldsymbol{q}_{ijk} \coloneqq \boldsymbol{b}_{\min} + \frac{(i, j, k)}{R} \odot (\boldsymbol{b}_{\max} - \boldsymbol{b}_{\min}).$$

Define $G_{ijk} \coloneqq f(\boldsymbol{q}_{ijk})$.



Example: sampling a level surface

Example (Scalar samples across an ellipsoid)

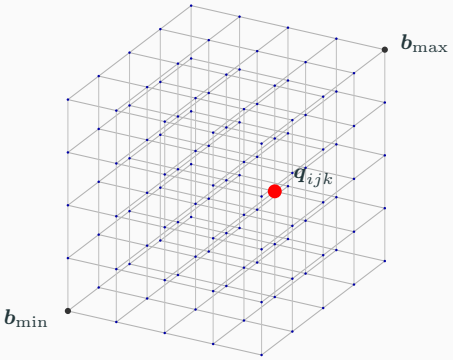
Use $f(x, y, z) = x^2 + 2y^2 + 3z^2$ and $\ell = \frac{4}{5}$.
Choose $\boldsymbol{b}_{\min} = (-1, -1, -1)$, $\boldsymbol{b}_{\max} = (1, 1, 1)$, and $R = 4$. Then

$$\boldsymbol{q}_{ijk} = \left(-1 + \frac{i}{2}, -1 + \frac{j}{2}, -1 + \frac{k}{2}\right).$$

For example,

$$G_{222} = 0 < \ell, \qquad G_{322} = \frac{1}{4} < \ell, \qquad G_{422} = 1 > \ell.$$

On the edge from $\boldsymbol{q}_{322} = (\frac{1}{2}, 0, 0)$ to $\boldsymbol{q}_{422} = (1, 0, 0)$, linear interpolation at level ℓ gives the point $(\frac{13}{15}, 0, 0)$.



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One standard method is **Marching Cubes** [4].

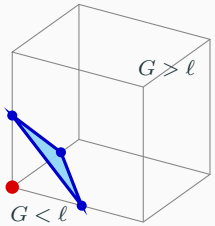
$$(\boldsymbol{V}, \boldsymbol{F}) = \mathrm{MarchingCubes}(\underline{\boldsymbol{G}}, \ell).$$

Surface extraction

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One standard method is **Marching Cubes** [4].

$$(\boldsymbol{V}, \boldsymbol{F}) = \mathrm{MarchingCubes}(\underline{\boldsymbol{G}}, \ell).$$

If only one vertex on the cube has a different sign, the face is a simple triangle as in the right figure.



Note: the surface extraction can be complicated

If multiple vertices on the cube has a different sign, the situation is quite complicated.

See, e.g.,
`https://www.boristhebrave.com/2018/04/15/marching-cubes-3d-tutorial/`.

Generating mesh from a field

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Question: How can we make the resolution of a 3D mesh adaptable?

Answer:

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Answer: We let a NN to output a **3D field**, instead of a 3D mesh directly. Then, we can apply a **surface extractor without trainable parameters** to generate a 3D mesh from the 3D field.

Outline

1. Introduction

2. Preliminaries: Mathematical Notations

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4. Adapter from a Latent Diffusion Model – Implicit 3D Field and Surface Extractor

5. The Complete 3D Rendering AI Pipeline – Hunyuan3D as an Example

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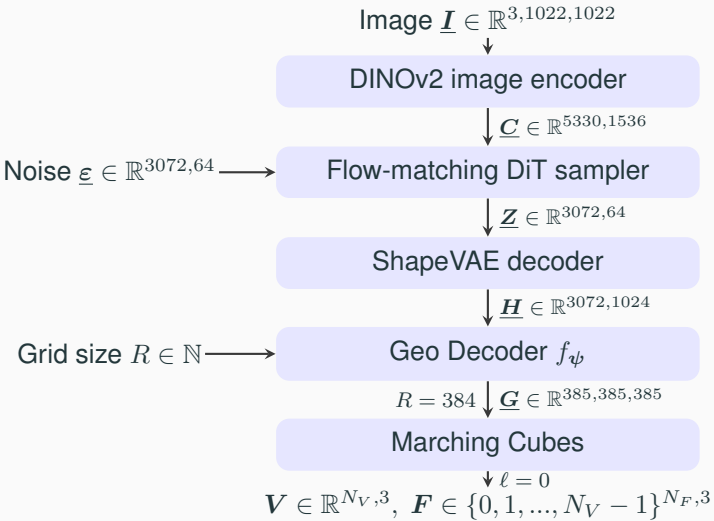
6. Summary

Question: How does a practical 3D rendering AI generate a 3D field and mesh?

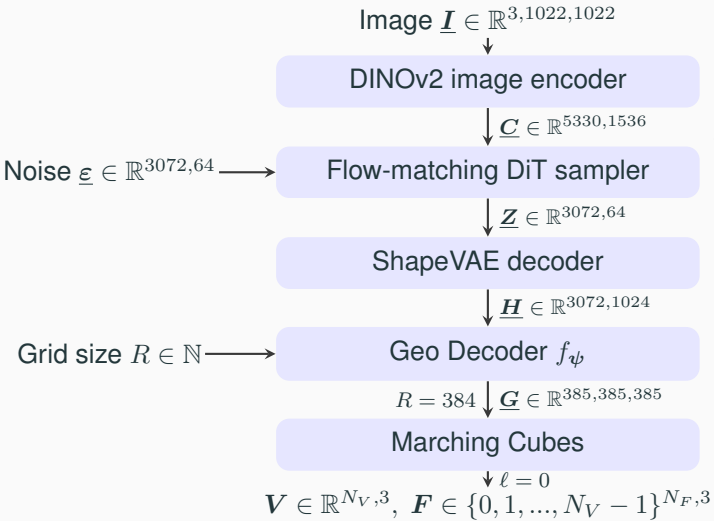
A concrete example: Hunyuan3D

We now use the shape-generation pipeline of Hunyuan3D-2mini Turbo as a concrete example [1, 3].

A concrete example: Hunyuan3D

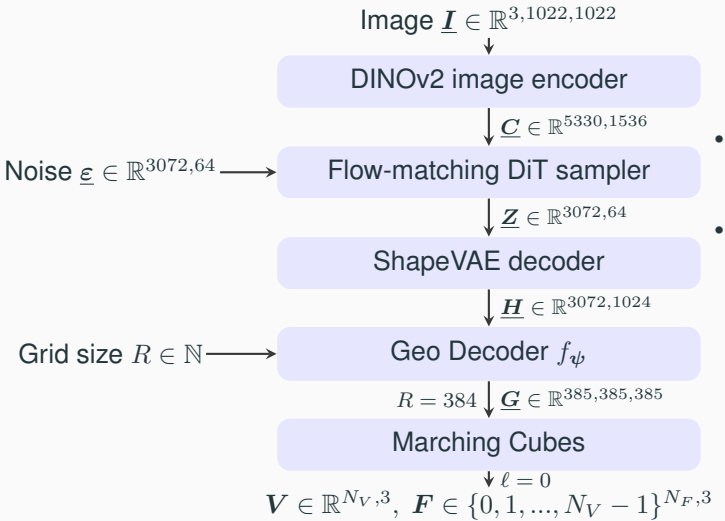


A concrete example: Hunyuan3D



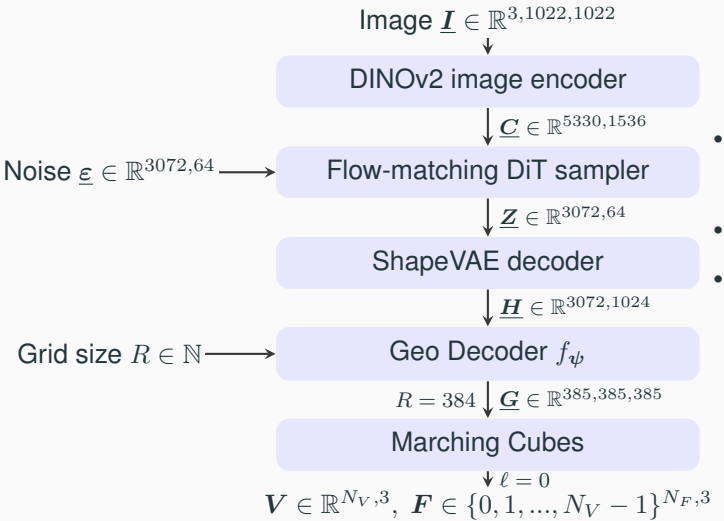
- The image is converted into a tensor $\underline{\boldsymbol{H}}$.

A concrete example: Hunyuan3D



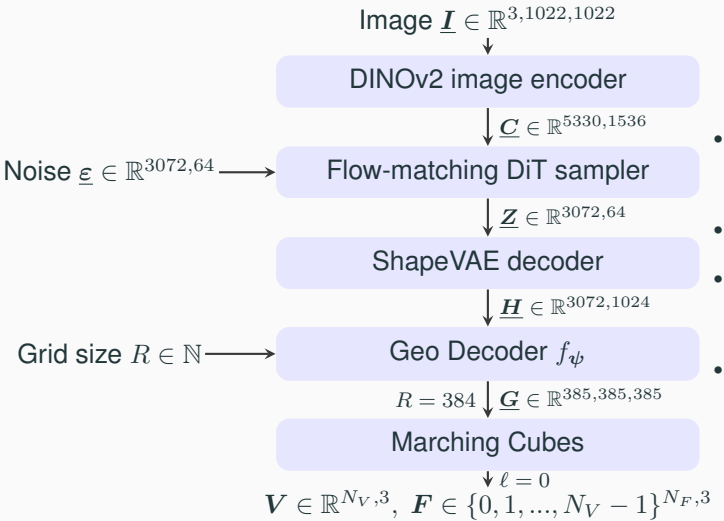
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- The image is converted into a tensor $\underline{\boldsymbol{H}}$.
- $f_{\psi}(\cdot, \underline{\boldsymbol{H}})$ defines a field.
- Values $\underline{\boldsymbol{G}}$ of the field are evaluated at the grid points.
- mesh are generated from the field value tensor $\underline{\boldsymbol{G}}$.

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- The goal of 3D rendering is to generate faces. Specifically, we are to generate a vertex matrix and a face matrix.
- We can construct faces from a 3D field.
- We let a NN generate a 3D field rather than generate faces directly. In this way, we can modify the resolution easily at the inference phase.

References i

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